Beyond Traditional Risk Scores: Tackling LS/CMI Offender Misclassifications with Machine Learning[*]

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Objectives. This paper investigates the accuracy of offender risk assessment scoring methods. We study the degree of misclassification resulting from the conventional practice of aggregating individual items to derive risk scores and categories. We document which types of offenders are prone to misclassification, particularly in relation to age and gender.

Methods. We use a machine learning algorithm to leverage the rich set of information available in the LS/CMI. Using all 45,535 assessments conducted between 2008 and 2015 in Quebec (Canada), we estimate probabilities from a random forest algorithm to predict individual risks of recidivism over a two-year follow-up. We compare the resulting probabilities to those inferred from the risk scores or categories to document the extent of misclassification. We devise a simple algorithm to construct alternative risk categories that reduce misclassification relative to the LS/CMI total scores and categories.

Results. The probabilities obtained from the random forest approach accurately predict individual probabilities to reoffend. Compared with these predictions, the traditional aggregation of items into risk scores or categories yields substantial misclassification for certain groups of offenders. In particular, we find that the risk associated with older individuals when using the LS/CMI risk categories is overestimated by about 10 percentage points. Our alternative risk categories, devised from our machine learning predictions, successfully avoid such misclassification.

Conclusions. Traditional methods of aggregating items from risk assessments into scores may lead to substantial misclassification, especially for older offenders. Misclassification arises from 1) items not being equally risk-relevant; 2) information collected by the LS/CMI being excluded or overly simplified when constructing scores; and 3) age being omitted from risk scores. Machine learning algorithms avoid these pitfalls and can be used to construct less biased categories.

Keywords: Risk assessment, Recidivism, Machine learning, LS/CMI

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1 Introduction

A number of offender risk assessment tools have been developed over the last 30 years, and especially so in North America. Canadian and American governments have promoted risk assessment research to guide offenders' supervision and ensure that decisions about their release are based on sound science, thus avoiding the pitfalls of misevaluating risk (Petrunik, 2002). Reliable and valid risk assessment tools benefit both convicted persons and the public. Overestimation of risk can lead to the long-term imprisonment of individuals who could otherwise be productive members of society, or impede their chances of reintegration upon release (Griffiths et al., 2007). Conversely, underestimation of risk can lead to the release of dangerous individuals and the (potential) perpetration of new crimes. Devising an unbiased risk assessment tool is a cornerstone of correctional practice (Brouillette-Alarie and Lussier, 2018).

Risk assessments are used to anchor the level and nature of services that will be offered to convicted individuals, as prescribed by the Risk-Need-Responsivity (RNR) model of Andrews and Bonta (2010). According to the risk principle, the level of supervision and service that is provided to convicted persons should match their risk level (Andrews et al., 1990; Andrews and Friesen, 1987; Kiessling and Andrews, 1980). According to the need principle, correctional interventions should focus on needs that are inherently criminogenic, since interventions focused on noncriminogenic needs are usually unable to reduce recidivism risk (Andrews, 1994; Hanson et al., 2009). Finally, according to the responsivity principle, correctional interventions should be further tailored to the personality, learning style, background, and beliefs of convicted persons (Andrews and Bonta, 2010). As such, risk assessments are used both to predict risk of recidivism and to guide rehabilitation efforts.

This paper focuses exclusively on the first role of risk assessment tools: predicting risks of recidivism. Risk assessment evaluations gather rich information on individuals, yet, not all the information is used to predict recidivism. Instead, this information is condensed in a simple score, which furthers determines a risk category (e.g. very low risk, low risk, etc.). This process involves numerous simplification rules and ignores potentially predictive information, which could result in misclassification of offenders' risk. To illustrate, consider a risk assessment tool with 40 questions, each worth one point. There are 137,846,528,820 unique ways to obtain a score of 20. Yet, despite none of these individuals having selected the same combination of responses, they would all be assigned the same predicted risk of recidivism. The problem becomes even more pronounced when these scores are further grouped into broader risk categories, where large intervals of scores are treated as equivalent.

The first contribution of this paper is to evaluate to what extent the oversimplified rules lead to misclassification of recidivism risk, compared to a generalized random forest approach (Athey et al., 2019) meant to exploit the available information fully. Our second contribution is to explore whether simple adjustments to the traditional practices would suffice to avoid misclassification. Finally, our third contribution is to show that we may construct new risk categories built from the machine learning predictions and avoid misclassification.

2 Background

2.1 The Use of Risk Assessment Tools and the Level of Service/Case Management Inventory (LS/CMI)

Over the years, evaluators have combined and weighted risk factors in different ways to assess the risk of criminal recidivism. According to Andrews and Bonta (2010), four generations of risk assessment tools may be distinguished. The first, unstructured clinical judgment, is characterized by a lack of constraints or guidelines for the clinician. The reliability and predictive validity of such tools are notoriously low (Dawes et al., 1989; Grove et al., 2000; Hanson and Morton-Bourgon, 2009; Meehl, 1954). This has prompted support for research on more structured methods of risk assessment. The second generation focuses on static actuarial assessments. These reliably determine the level of risk by mechanically combining empirically validated predictors. Each item is associated with a specific weight and coding rules are very specific, thus leaving little room for professional input. The risk score corresponds to the sum of the separate items. Each value of the score is associated with an expected likelihood of recidivism and/or risk category. The first actuarial tools were based on static risk factors only and did not account for changes in convicted persons' experience. Consequently, they had limited clinical relevance (Bonta, 1996, 2002; Douglas and Skeem, 2005; Hanson and Harris, 2000; Gendreau et al., 1996). Third-generation instruments include both static and dynamic risk factors in assessing the risk of reoffending. Though dynamic risk factors require more clinical expertise to score than static ones, coding rules of third-generation tools are very specific (Andrews and Bonta, 2010; Bonta, 1996). Fourth-generation scales, case management risk/need tools, expands on the clinical relevance of third-generation instruments by providing clear guidelines to ensure that case management is consistent with the results of risk assessment in accordance to the risk and need principles. In addition, case management tools assess protective factors and ensure adherence to the responsivity principle by considering clinically relevant factors such as the learning style and personality of convicted persons (Andrews and Bonta, 2010).

Case management risk/need tools are relatively novel. Consequently, few such tools are currently available. To date, only the *Level of Service/Case Management Inventory* (LS/CMI: Andrews et al., 2000) has received substantial validation. The tool has been validated for men, women, adults, adolescents, both incarcerated and parolees.[1] The analysis in this paper focuses on this tool. It comprises eight risk domains that have all demonstrated validity in predicting recidivism (Andrews and Bonta, 2010; Andrews et al., 2011; Giguère et al., 2023; Olver et al., 2014). The LS/CMI items are provided in Table A.1. After the evaluation is conducted, each of the 43 item is given a score of 0 or 1, and all items are aggregated to obtain a score between 0 and 43. This score determines the risk category assigned to the individual: very low risk (0-4), low risk (5-10), medium risk (11-19), high risk (20-29), or very high risk (30-43). We provide more details about what is observable in the evaluation in the Data section.

2.2 Limitations of Traditional Models of Risk Assessments

Criminological risk assessment tools like the LS/CMI have been developed based on traditional statistical models, which have been found to have several limitations. First, they assign equal weights to all items when computing an aggregate score. This implicitly assumes that the items all have the same predictive power. This assumption has been challenged by item-response theory studies of risk tools such as the LS/CMI (Giguere and Lussier, 2016; Giguère et al., 2023) or Static-2002R (Brouillette-Alarie et al., 2023), which found substantial variation in individual items' ability to discriminate between offenders' types, suggesting that attributing the same weight to each item when computing the score might not be optimal.

Second, by simply aggregating the items, risk tools omit potentially important interaction effects. For example, studies have found that the effect of protective factors can be best understood as factors that reduce risk when interacted with already existing risk factors (Guay et al., 2020). Third, potentially relevant information, such as age and

[1] For reviews, see Andrews and Bonta (2010) and Olver et al. (2014).

gender, is often omitted when computing risk scores. Such omissions may introduce measurement errors and thus bias individual risk scores in one way or another. Specifically, the omission of age in the LS/CMI is quite surprising considering the numerous studies that found that adding age items to criminological actuarial scales significantly improved their predictive validity (Helmus et al., 2012; Wollert et al., 2010).

More generally, rich information is left unexploited from the LS/CMI and other risk assessment evaluations. As indicated in Table A.1, which describes the items of the LS/CMI, many items are initially measured on a scale of 0 to 4 and then binarized, while other sub-items are simply discarded when computing the scores. For instance, while Item 1 captures whether the individual has any prior convictions, sub-items 1.1 and 1.2 record the number of previous youth dispositions and adult convictions, respectively. Similarly, Item 7 is a simple indicator of past institutional misconduct, yet sub-item 7.1 documents the number of behavioral infractions during prior sentences. These sub-items (1.1, 1.2, and 7.1) are disregarded in the scoring process.

2.3 Machine Learning Methods to Predict Recidivism

Some of the limitations discussed in the previous section—such as the equal weighting of items and the omission of key predictors like age—could be addressed using traditional statistical methods. For example, a logistic model would estimate the importance of each item through its coefficients and allow for the inclusion of additional predictors such as age and gender. However, this approach may still leave a substantial amount of information unexploited, as traditional models require the imposition of a specific functional form and may struggle to fully capture complex interactions or nonlinear relationships between risk factors.

Machine learning (ML) algorithms are well-suited to exploit the features of the LS/CMI to the fullest. One of the primary benefits of ML is its ability to process large amounts of data, including when variables are highly correlated, and to identify patterns and relationships that may be difficult or impossible to identify using traditional statistical methods. ML can efficiently incorporate all relevant information in predicting recidivism probabilities, intrinsically accounting for interaction effects between all factors and weighing individual factors optimally. In particular, random forest ML algorithms are designed to capture interactions and non-linearities while avoiding overfitting (Bilger and Manning, 2015). In the criminal justice space, they have been found to lead to fairer estimates, which could contribute to reducing racial disparities in release rates and avoid unnecessary denials of parole (Laqueur and Copus, 2022; Wang et al., 2023).

Several studies have leveraged machine learning methods to predict criminal recidivism (Berk et al., 2006; Berk and Bleich, 2014; Laqueur and Copus, 2022; Wang et al., 2023). For instance, Duwe and Kim (2017) conducted a comparative analysis of the performance of twelve supervised learning algorithms and found that the *LogitBoost* algorithm yielded the highest accuracy. Travaini et al. (2022) consolidated the findings from twelve distinct studies that employed a variety of models, including logistic regressions, random forests, and boosting. They observed that these algorithms could typically achieve an accuracy of approximately 80% and an area under the curve[2] (AUC) of 0.74. Most relevant to the current research is the work of Ghasemi et al. (2021) who trained decision trees, random forests, and support vector machines on roughly 100,000 LS/CMI evaluations. Their results showed that the support vector machine exhibited the highest predictive validity, with an AUC of 0.7545. However, this method marginally outperformed the standalone LS/CMI scores, which had an AUC of 0.7517.

Considering these advantages, the current study applies ML random forest algorithms to data from LS/CMI evaluations of convicted individuals evaluated in Quebec (Canada) correctional facilities between 2008 and 2015. The ML algorithm processes all available relevant information collected when administering the LS/CMI tool, including information not used to compute standard LS/CMI scores. This allows us to compute the best possible individual-level recidivism probabilities based on the entirety of available information. By comparing these to the prediction based on risk categories that results from a simple aggregation of items, we can measure the extent of misclassification associated with traditional tools. In line with the LS/CMI, we propose alternative risk categories based on ML predictions that are not only highly predictive of recidivism but also designed to be easily integrated into current practices with minimal adjustments. Differing from past studies employing classification methods on criminological risk scales, our study places emphasis on the generated probabilities rather than the predicted classes, allowing us to measure the level of risk itself (see Wang et al. (2023) for a similar application).

[2]The area under the curve refers to a measure of predictive accuracy based on a tool's ability to distinguish between individuals who do and do not reoffend. A value of 0.5 indicates that the predictive accuracy of the risk assessment tool is equivalent to random chance.

3 Data

Our analysis is based on the Administrative Correctional Files (DACOR)[3] of the Quebec Ministry of Public Security, Canada. The Ministry is responsible for managing all offenders serving sentences of less than two years.[4] We use two data files from DACOR: the LS/CMI evaluation file and the sentence file.

The LS/CMI evaluation file provides detailed information on all provincial offenders who are sentenced for at least six months, whether they are incarcerated or serving a community sentence. Offenders sentenced for less than six months are not assessed with the LS/CMI. As stressed above, the LS/CMI tool contains three additional items that are not included when computing the risk score but that are available in our data. Furthermore, while most item responses are binary indicators, some use a 4-point scale. These are converted into binary indicators for the purpose of computing the risk score. We observe the original 4-point value. The LS/CMI further documents the gender and date of birth but omits this information from any scoring purposes. In order to retain the maximum amount of information when establishing our benchmark predictions, our analysis uses the 4-point scales rather than their dichotomized version when applicable, comprises items 1.1, 1.2, 7.1, and integrates age and gender items.[5]

The sentence file records all sentences that were pronounced between 2008 and 2020. It spans up to five years beyond the period covered by the LS/CMI file (2008-2015). We define recidivism as the occurrence of a new sentence after release within a two-year follow-up period starting at the planned day of release of the current sentence.[6]

We match the LS/CMI records with the sentence records (and recidivism measures) using individual-level identifiers. Our resulting sample includes 45,535 evaluations con-

[3]In French: Dossiers Administratifs CORrectionnels.

[4]In Canada, offenders with sentences of two years or more are incarcerated in a federal prison, and thus are excluded from our data.

[5]See Table A.1 for detailed summary statistics.

[6]Even if all the evaluations are conducted in provincial facilities, federal sentences are also recorded in the sentencing file. Thus, our recidivism measure include both provincial and federal charges.

ducted over 25,629 unique offenders.[7] The sample is predominantly male (93%), and the average age is 35.37 (SD = 11.58). The average total risk score is 25.30 (SD = 8.26), and 52% of offenders are found to have at least one subsequent conviction within the follow-up period.

4 Methodology

4.1 Overview

We seek to measure the extent of misclassification that results from reducing the rich information available from LS/CMI evaluations to simple risk categories. We proceed in six steps:

1. We use a random forest machine learning algorithm to obtain individual-level probabilities of recidivism based on all available information in the LS/CMI evaluations. These probabilities serve as a benchmark against which we evaluate simpler and more interpretable approaches. We elaborate on this method below. We also examine which elements of the LS/CMI evaluation contribute most to predicting recidivism risk.

2. Next, we investigate how well the LS/CMI risk categories (*i.e.* very low, low, medium, high, very high) predict recidivism. To do so, we compute the average recidivism rate for each category.

3. We compare the ML probabilities to those of the LS/CMI risk categories. This allows us to measure the misclassification that results from using the simple categories, compared to exploiting all available information using ML algorithms. We

[7] In traditional regression models, correlated observations—such as multiple evaluations for a single individual—can be accounted for by clustering the standard errors, accommodating the within-group correlation and adjusting the standard error estimates. However, machine learning algorithms lack a direct counterpart to this method. Instead, to counteract potential biases from such non-independent observations, we split the dataset into distinct training and test sets, where all the evaluations pertaining to a certain individual fall into one of the two groups. Following Grogger et al. (2021), we randomly selected 10% of the individuals (2,563) to form a test sample. Consequently, all assessments associated with an individual are either in the training set ($N = 40,974$) or the test set ($N = 4,561$), and we evaluate all models' performance using only the test sample. We also employed a cluster-robust forest, wherein clusters are defined by individuals. Instead of drawing individual evaluations at each iteration, entire clusters (i.e., all evaluations of a particular individual) are sampled. The results are identical to prior findings. The detailed results from this approach are available upon request.

then investigate which demographic groups (*i.e.* by age and gender) are most at risk of misclassification.

4. We explore how simple adjustments to the LS/CMI can reduce misclassification. We compute the predicted probabilities from using:

   (a) the precise LS/CMI score,

   (b) a logistic regression using the 43 items as predictors,

   (c) a logistic regression using the 43 items as predictors, adding age and gender.

5. Because using individual-level probabilities may not be convenient in practice, we solve a simple minimization problem to construct risk categories that minimize the risk of misclassification relative to the ML probabilities. We also compare the performance of these new categories to the other methods described above.

4.2 Random Forest

We estimate individual-level probabilities of recidivism based on all available information in the LS/CMI, adopting a probabilistic approach rather than the traditional classification methods often used in machine learning. Unlike traditional classification, which predicts a label (e.g., "recidivates" or "does not recidivate") by estimating a probability and assigning a label when a threshold (typically 50%) is crossed, probabilistic classification treats the probability itself as the outcome of interest (Kruppa et al., 2014), which is critical for classifying offenders by risk level.

Although some limitations of the LS/CMI—such as the arbitrary weighting of items and the omission of age—can already be addressed with logistic regressions, machine learning methods offer additional advantages. In particular, random forests provide a flexible framework to fully exploit the information available in the LS/CMI evaluations. Unlike logistic regressions, which assume a linear relationship between predictors and outcomes, random forests allow for nonlinearities and interactions, mitigating issues related to multicollinearity by automatically selecting and weighting the most relevant features (Ranganathan et al., 2017). Furthermore, random forests are designed to reduce overfitting when dealing with several interactions (Bilger and Manning, 2015). Interestingly, random forest algorithms have been shown to outperform classification trees and logistic regressions in predicting prison misconduct (Berk et al., 2006).

We use the generalized random forest approach developed by Athey et al. (2019) and the accompanying package (Tibshirani et al., 2021). Within this framework, the algorithm constructs decision trees by recursively partitioning the data into smaller subsets. At each node, the algorithm selects a predictor and a split point that minimizes the Gini coefficient, thereby maximizing the model's ability to distinguish between those who recidivate and those who do not. The tree continues to grow until further reductions in Gini impurity are no longer possible or until a stopping criterion is met (in our case, a minimum leaf size of five). Two important features of the Athey et al. (2019) algorithm are worth highlighting further: bagging and honest splitting.

Bagging: Each tree is constructed using a bootstrapped sample of the training data, which leaves some observations out. These unused observations, known as out-of-bag (OOB) samples, are used for generating unbiased in-sample predictions. Predictions for each data point are made using only the trees where that point was not included in the training set.

Honest splitting: To further reduce overfitting, half of the in-bag observations are used to determine the splitting criteria, while the remaining 50% are reserved for populating the tree and estimating recidivism rates in each leaf. This method helps ensure that the model generalizes well to new data.

In line with standard random forest practices, we allow for feature sampling, selecting a subset of $\sqrt{p} + 20$ variables at each split, where p is the number of available predictors. This tree-building process is repeated 2,000 times, each time using a different sample of offenders to train the model and generate predictions. We train the random forest using 90% of the dataset, reserving 10% for testing the external validity of the algorithm.

4.3 Simple Adjustments and Misclassification

In addition to the ML probabilities discussed above, we compute probabilities from four other methods. Firstly, we compute the predicted probabilities from using the simple LS/CMI categories. To do so, we simply compute the recidivism rates associated with each category. From there, we define our measure of misclassification. We define the extent of misclassification by measuring the difference between the probability of individual i's risk category (or, alternatively, within their score) and the probability predicted by the ML algorithm. Specifically, let the error for individual i be given by

$$\text{error}_i = \bar{p}^{\text{ category}_i} - \hat{p}_i^{\text{ML}}, \tag{1}$$

where $\bar{p}^{\,\text{category}_i}$ represents the average probability of reoffending for those classified in a specific LS/CMI risk category, while $\hat{p}_i^{\text{ML}}$ is the predicted probability provided by ML. A positive error indicates that the LS/CMI risk is overestimated while the converse holds for a negative one.

Secondly, we compute probabilities that take into account the precise LS/CMI total scores. Specifically, we compute recidivism rates for each of the 43 possible LS/CMI total scores and define the associated misclassification error in a manner similar to Equation (1).

Thirdly, we test some simple adjustments to the traditional LS/CMI by departing from the aggregation procedure. We estimate logistic regressions of the form

$$P(recidivism_i = 1) = F\left(\sum_{j=1}^{43} \omega_j \mathbb{1}\{item_{ij} = 1\} + \mathbf{x}_i'\boldsymbol{\beta}\right),$$

where F is the cumulative logistic distribution and where $\mathbf{x}_i'$ contains a constant and, depending on the specification, the age and gender of the offender. In a manner identical to Equation (1), we define a misclassification measure using the predicted probabilities generated from these regressions.

4.4 Defining New Categories

For practical reasons, case management is better achieved with risk categories than with individual probabilities. To generate risk categories from our ML probabilities, we propose a simple minimization algorithm to determine probability cutoffs to mimic risk categories. In keeping with the LS/CMI framework, we define five new risk categories in the application. However, this approach is flexible and can easily be extended to accommodate any number of categories, as described below.

The general problem can be written as follows. Let $p_1, p_2, \ldots, p_n$ be the data points (i.e., the probabilities provided by ML) that need to be categorized, and let $c_1, c_2, \ldots, c_{K-1}$ represent the cutoff points that divide the data into K categories. The objective is to minimize the sum of squared errors (Equation 1) between the data points in each category and the mean of that category.

The minimization problem is given by:

$$c_1, c_2, \ldots, c_{K-1} = \arg\min \sum_{k=1}^{K} \sum_{i \in C_k} \left(p_i - \bar{p}^k\right)^2, \tag{2}$$

where

$$\bar{p}^k = \frac{1}{|C_k|} \sum_{i \in C_k} p_i$$

is the mean probability in category k. C_k is the set of indices i such that p_i falls into category k. More precisely, the categories are defined as follows:

- Category 1: $p_i \leq c_1$
- Category 2: $c_1 < p_i \leq c_2$
- ...
- Category $K-1$: $c_{K-2} < p_i \leq c_{K-1}$
- Category K: $p_i > c_{K-1}$.

Figure 1 visually represents the process of dividing the probability space into K categories. The circles (points) represent individual probabilities p_i. The small vertical lines mark the cutoffs $c_1, c_2, \ldots, c_{K-1}$, which define the boundaries between categories. The error is the difference between the actual probabilities p_i and the average probability in their category. To illustrate, the average probability for the middle category is shown as a larger circle. The aim is to create categories such that the probabilities within each category are as close as possible to the average of that category, minimizing the overall variance within each group.

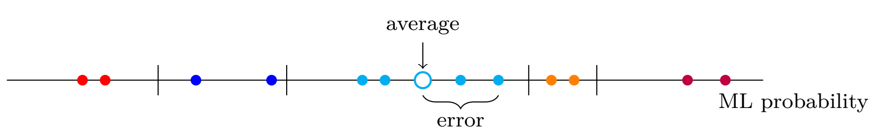

Figure 1: Description of the Minimization Problem

Notes: The figure shows the intuition behind the minimization problem used to define the new risk categories. The goal of the algorithm is to find the cutoffs that minimize the (squared) distance between the individual probabilities and the average probability within a category.

To solve the minimization problem (2), we use the Nelder-Mead optimization method, starting from the quintiles of the distribution as the starting points.[8]

5 Results

5.1 Predicted Probabilities of Recidivism

The predicted individual probabilities of recidivism obtained with the ML algorithm are shown in Figure 2 along with confidence intervals. The figure ranks the individuals in increasing order of predicted probability in the test sample.

Unsurprisingly, since these probabilities are predicted using all information available in the LS/CMI evaluation, their predictive power outperforms those of standard LS/CMI metrics, both in- and out-of-sample. This is confirmed in Table 1, which reports three performance measures for the ML predictions (in dark-shaded cells) and for all our other prediction methods discussed in Section 4. The rows present each model in order of complexity, starting with the simplest and progressing to those incorporating the most information. The columns report, for the test sample:

- the correlation between predicted probabilities and actual recidivism,
- the Brier score[9], and
- the area under the curve (AUC) of ROC curves.

The out-of-sample correlation coefficient for the ML algorithm is 0.4546, outperforming the correlation coefficients of both the risk categories (0.3883) and the total scores (0.4043). Similarly, the ML algorithm achieves a higher out-of-sample AUC of 0.7611, compared to 0.7115 for the risk categories and 0.7318 for the total score. The last two columns report p-values from DeLong et al. (1988), which assess whether differences in AUC between models are statistically significant. The second-to-last column tests whether each incremental improvement (for example, adding age and gender to the logit regression) significantly increases the AUC compared to the previous model. The final column

[8]While a grid search could also be used to find the optimal cutoffs, it is computationally intensive. The cutoffs obtained with a grid search method are almost identical to those derived from the Nelder-Mead method. These results are available upon request. Additionally, both methods allow for the inclusion of a penalty term to ensure that the solution always results in the correct number of categories, avoiding any empty ones during the optimization process.

[9]The Brier score is defined as $N^{-1}\sum_i (p_i - y_i)^2$, where p_i is the estimated probability and y_i the actual outcome. It thus ranges between 0 and 1, with 0 indicating perfect predictions. It is equivalent to the mean squared error as applied to predicted probabilities.

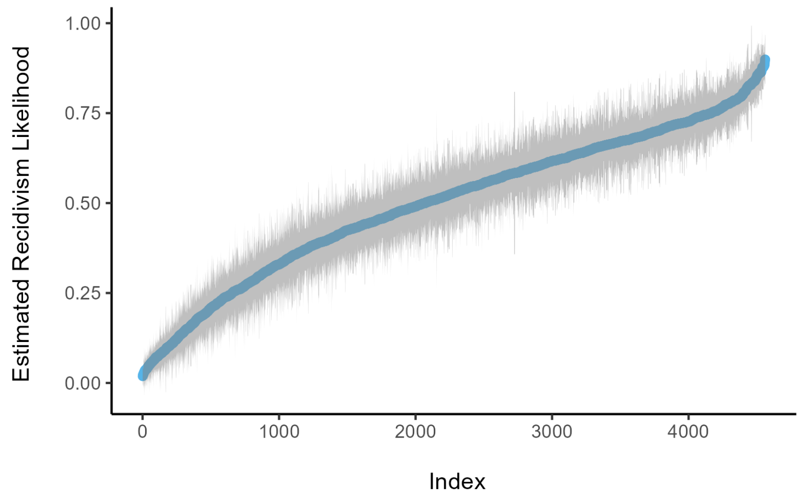

Figure 2: Individual Probabilities and 95% Confidence Intervals

Notes: The figure shows the estimated probabilities of recidivism and the associated 95% confidence intervals in the hold-out sample.

compares each model's AUC to that of the machine learning model. The AUC from the ML-based probabilities is statistically significantly higher than those from the risk category and total score models.

5.2 Which Components of the Evaluation Matter Most?

The estimated probabilities are based on a wide array of factors, including age, gender, and specific sub-items of the LS/CMI. It is natural to ask which of these variables have the most predictive power. The relative importance of each variable is determined by quantifying how frequently it is selected by the machine learning algorithm to separate recidivists from non-recidivists (Archer and Kimes, 2008).[10] Figure 3 depicts the importance measures of the 15 most significant variables. The variables are classified into two categories: gray labels are included in the calculation of standard LS/CMI scores, while blue labels are usually collected during the evaluation but not directly used in the scoring.[11]

[10]More precisely, the importance of a variable is proportional to the number of times it was selected by the algorithm to create two nodes. Variables chosen nearer the top of a tree are assigned greater importance, as proposed in Tibshirani et al. (2021).

[11]Table A.1 in the appendix provides a description of the LS/CMI labels.

Table 1: Models Summary (out-of-sample)

	Correlation	Brier Score	AUC		
	metric	*metric*	*metric*	*p-value: across models*	*p-value: vs ML*
Risk Category	.3883	.2117	.7115 (± .0140)		< .01
				< .01	
Risk Score	.4043	.2085	.7318 (± .0144)		< .01
				< .01	
Logit with 43 items	.4358	.2019	.7487 (± .0141)		< .01
				.47	
Logit with 43 items + Age + Gender	.4403	.2010	.7501 (± .0141)		< .01
				.62	
New Risk Category	.4443	.2011	.7484 (± .0137)		< .01
ML	.4546	.1990	.7611 (± .0138)		

Notes: The table shows accuracy metrics between recidivism and the probabilities of recidivism obtained from the risk category, total score, a logistic regression on the 43 items, a logistic regression that adds age and sex, and the machine learning algorithm. The metrics considered are the coefficient of correlation, the Brier score (defined in the text), and the area under the curve (AUC, with the 95% margin in parentheses). The metrics are presented for the out-of-sample group (4,561 individuals).

Certain items play a much larger role than others in improving the accuracy of the estimated probabilities. For instance, Item 8, which indicates whether a charge was laid, probation was breached, or parole was suspended during prior community supervision, stands out with an importance score nearly three times higher than the second most important item. In contrast, most items (including those not shown in the figure) contribute very little to the overall predictive power. This finding suggests that the equal weighting of items in the standard LS/CMI scoring system is suboptimal.

Another key insight is that some highly predictive information is collected during the evaluation but not used in the standard scoring. For example, Item 7.1—the number of prior behavioral reports, which ranges from 0 to 10 in our dataset—is critical for classifying risk. Yet, in the standard LS/CMI scoring, this sub-item is reduced to a simple binary measure (institutional misconduct), which ranks as the third most important item.

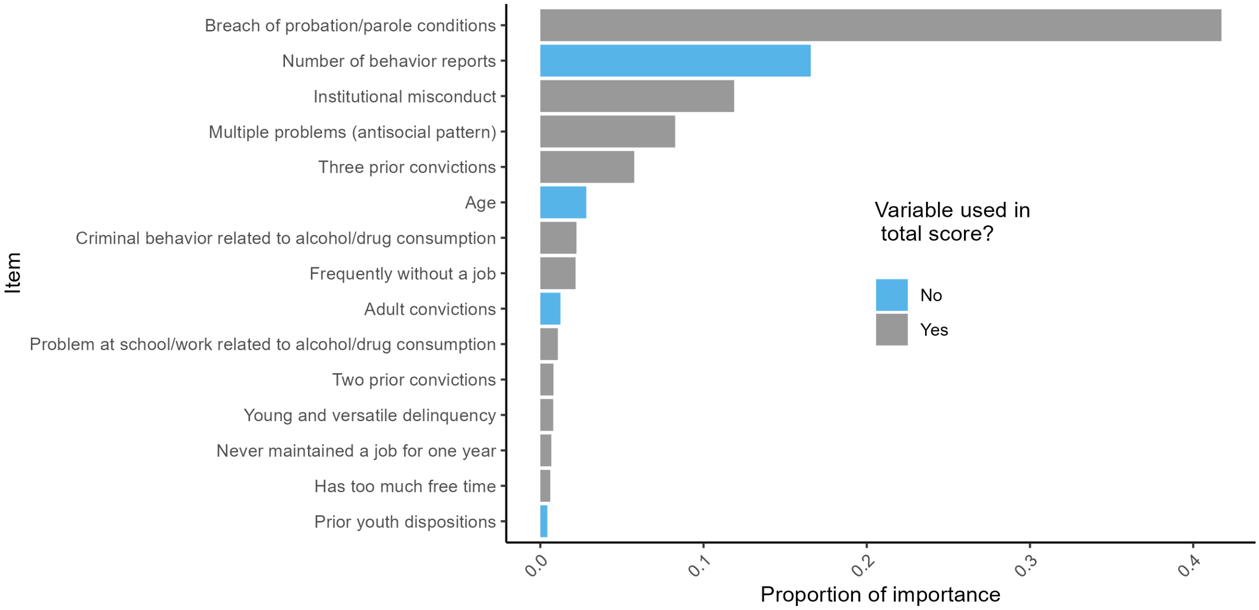

Figure 3: Variable Importance (Truncated—15 most important variables shown)

Notes: The relative importance of each variable is determined by quantifying how frequently it is selected by the machine learning algorithm to separate recidivists from non-recidivists. Items in gray are included in the computation of the LS/CMI risk scores. Item in blue are excluded from the risk score. Only the 15 most important items are shown in the figure.

Additionally, age emerges as a significant predictor of recidivism; though it is recorded in the LS/CMI, it is not factored into the risk score. Interestingly, the analysis reveals that gender, also documented in risk assessments, is not a primary predictor of recidivism once the full range of LS/CMI data is taken into account. Gender ranks 43 out of 48 variables, with an importance score of just 0.000152, meaning it contributes to only about 0.02% of the decision splits.

5.3 Misclassification

As stressed above, assessing an offender's risk of recidivism on the basis of broad risk categories defined over an aggregate risk score may result in considerable misclassification. To illustrate, Figure 4 draws the box plot distribution of the estimated ML probabilities for individuals according to their LS/CMI risk category. Several results are worth emphasizing. First, as we move from *very low* to *very high* risk categories, the median probability of recidivism reported by the lines in the box plots increases as well. There is thus a strong positive correlation between the LS/CMI risk categories and the estimated probabilities of recidivism.

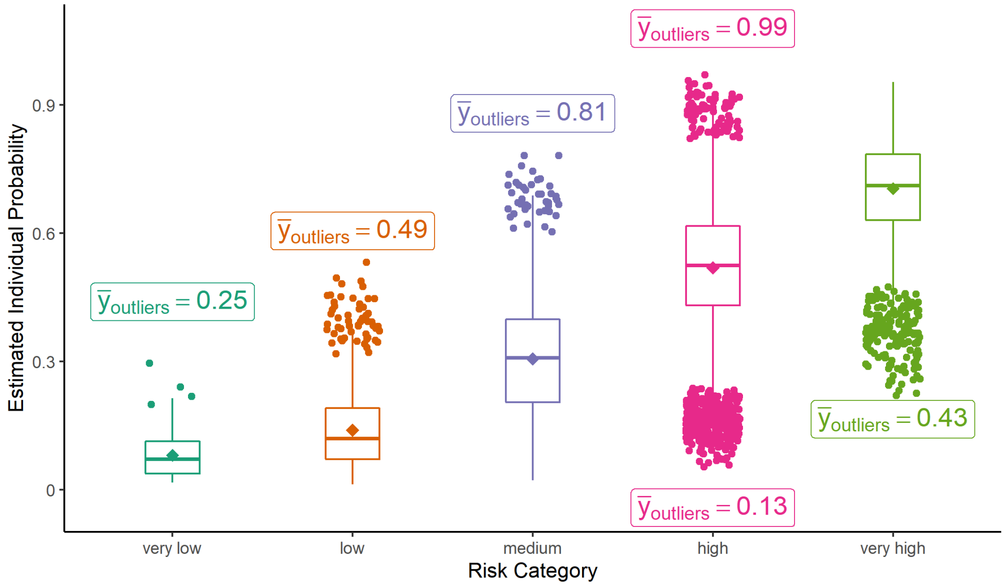

Figure 4: Distribution of Individual Probabilities by LS/CMI Risk Category

Notes: The figure illustrates the distribution of individual probabilities of recidivism, as estimated by machine learning (ML), conditional on the LS/CMI risk category. Dots represent individuals identified as outliers. An outlier is defined by a probability to recidivate that is either lower than the first quartile minus 1.3 times the interquartile range or higher than the third quartile plus 1.3 times the interquartile range. The rectangular boxes show the actual recidivism rates corresponding to these outliers. Inside the box plot, the lines represent the median predicted probability, and the square represents the average predicted probability.

Second, and most interestingly, the ML algorithm is able to identify numerous outliers for each LS/CMI risk category. Above and below the box plot, we report the empirical recidivism rates of the outlier observations depicted as colored dots. For instance, 49% of the outlier individuals in orange recidivated within two years. Some individuals deemed at very low risk (*i.e.* whose LS/CMI risk score ranges between 0 and 4) have relatively high probabilities of recidivism according to the ML algorithm. Indeed, these outliers have an actual reoffending rate of over 25%, way above the average of their category. Yet, the dots (probabilities predicted by our ML algorithm) on the y-axis align almost perfectly with this actual recidivism rate. Likewise, outliers deemed at *low risk* (LS/CMI risk score between 5 and 10) have an actual recidivism rate of 49%, more than twice as high as the average of their category. Again, the predicted probabilities coincide with their observed rates. Finally, among those deemed at *high risk*, one outlier group has a reoffending rate as high as 99% while the other group recidivates at the rate of 13%, very close to their probabilities of recidivism predicted by our ML algorithm.

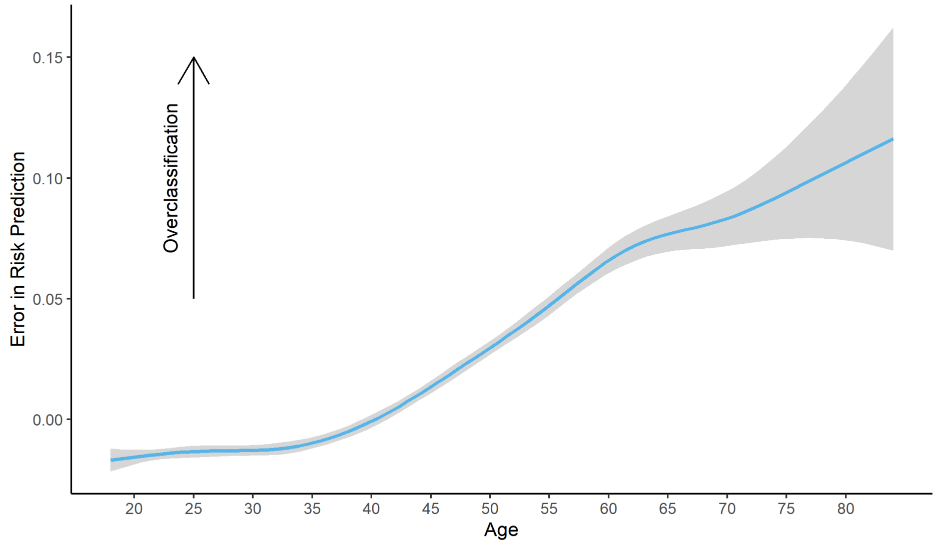

Figure 5: Errors in Risk Predictions Based on Age

Notes: The error in prediction is defined as the difference between the probability given by the risk category and the probability obtained via ML. A positive error indicates that the risk category overestimates the risk of recidivism; in contrast, a negative error means an underestimation. The plot shows that the error in prediction tends to increase with the individual's age.

Next, we use a regression tree[12] to investigate whether external characteristics predict these errors. Specifically, we examine whether gender or age at sentencing correlates with the errors calculated with Equation (1). The tree is depicted in Figure A.1 in the Appendix. It shows that age is the most important determinant of misclassification, with older individuals being most likely to be overclassified. Figure 5 plots the relation between the errors and age. Overestimation of risk increases systematically with age. Individuals above 40 years of age are substantially overclassified, some by as much as 10 percentage points. Interestingly, the analysis from the regression tree revealed little influence of gender on classification errors.[13]

5.4 Can Simple Adjustments to the LS/CMI Reduce Misclassification?

The simplest adjustment to the LS/CMI would be to weigh items based on their predictive power. To this end, we estimate logistic regressions of recidivism of the 43 items of the LS/CMI, and add gender and age in a linear fashion. As shown in Table 1, the estimates

[12]A regression tree is a decision tree model used for predicting a continuous value (the predicted errors in our setting).

[13]Figure A.2 illustrates that the error distributions by gender are almost confounded.

of risk correlate well with recidivism, more so than the risk category and risk score, but slightly less than the ML algorithm.

To investigate how the alternative methods described in Section 4 can reduce misclassification, we compute the classification errors arising from each estimation method by comparing their predicted probabilities to those from the ML algorithm (similar to Equation 1). The upper panel of Figure 6 presents the distribution of prediction errors using the LS/CMI categories. The distribution is centered around zero, indicating that LS/CMI categories do not exhibit a systematic bias towards either overestimation or underestimation of risk. However, the substantial spread of the distribution underscores the considerable likelihood of misclassification, indicating that errors occur frequently but without a systematic directional bias. The standard deviation of errors is around 0.13, so that, on average, an individual will be predicted a probability of recidivism that is more or less 13 percentage points different from the value he would have using an ML probability.

As shown in the second panel, relying on the precise LS/CMI score as opposed to the risk category still results in misclassification, although the distribution of errors displays slightly less variation. Using a logistic regression with the 43 binary items (third panel) further reduces the spread of misclassification errors. Finally, including age and gender, in addition to the items, in a logistic model reduces misclassification slightly more, resulting in a standard deviation of errors of 0.083. Overall, each of these simple adjustments can reduce misclassification, though the variance of the errors is still sizable in every case.

5.5 Defining New Risk Categories to Minimize Misclassification

While the ML predictions are the most predictive of recidivism, using individual-level probabilities may not always be practical. To address this, we solve the minimization problem outlined in Section 4.4 to create new risk categories that reduce misclassification errors. The five resulting categories, referred to as *New Risk Category* in Table 1, have cutoffs at 0.24, 0.43, 0.58, and 0.73, which were determined to optimize predictive performance. These categories exhibit high predictive accuracy, comparable to the logit regressions, though slightly lower than the ML algorithm. This suggests that grouping ML probabilities into categories leads to minimal information loss, especially when contrasted with the original risk categories based on simple aggregated items.

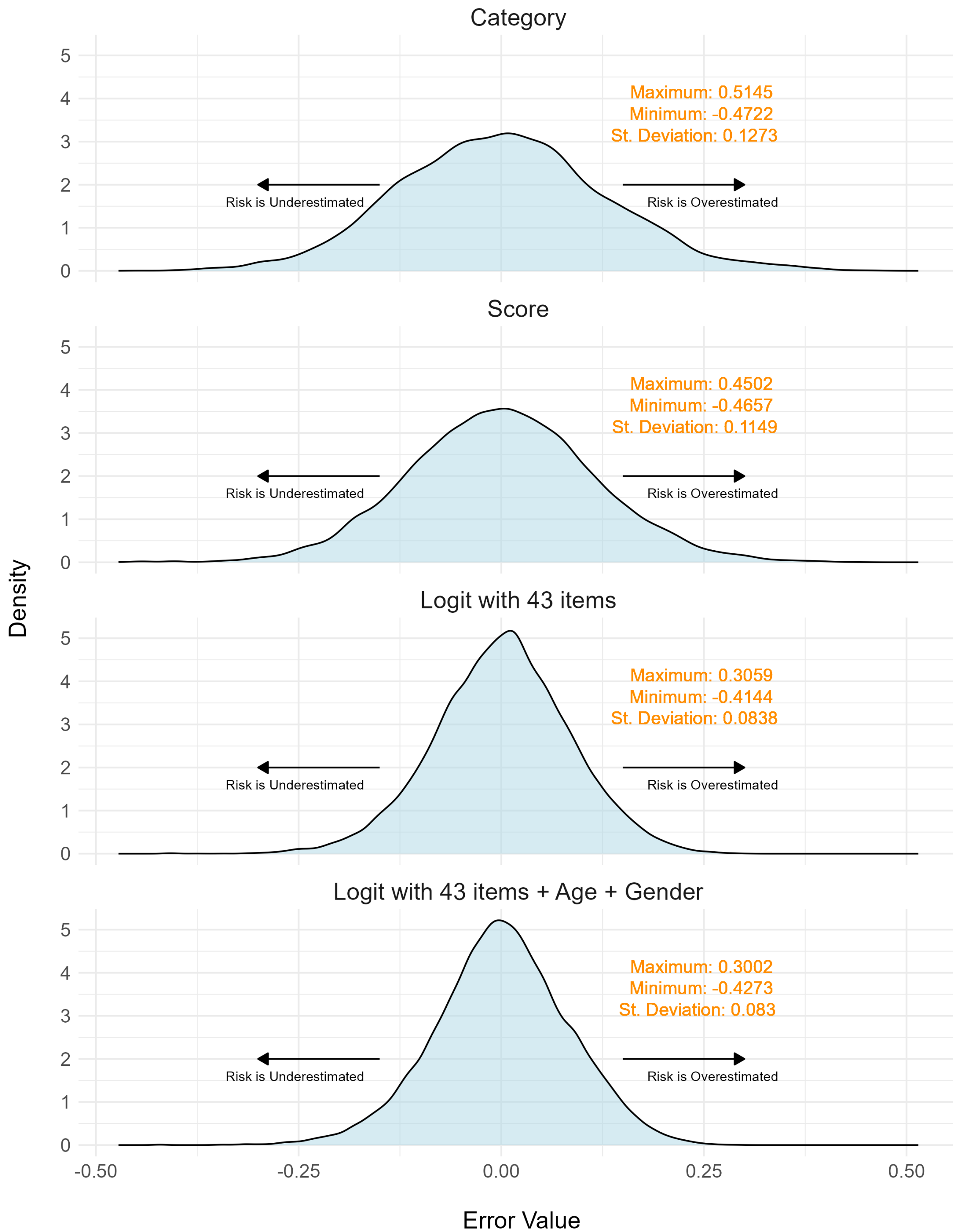

Figure 6: Error Densities

Notes: The error in prediction is defined as the difference between the probability given by an aggregation rule (risk category, score, weighted score with or without an age effect) and the probability obtained via ML. A positive error indicates that the risk category overestimates the risk of recidivism; in contrast, a negative error means an underestimation. The plot shows that the error in prediction tends to decrease as more refined methods are used.

6 Discussion

The current study measures the extent of misclassification resulting from the aggregation of LS/CMI risk scores into categories and shows how ML algorithms can improve risk evaluations. When applied to data collected from traditional risk assessments, ML algorithms can 1) assign different weights to risk factors to account for their predictive power, 2) take into account interaction effects between items, and 3) easily take into account information collected within the risk assessment but not computed in risk scores/categories (e.g., age, gender, sub-items). Comparing the ML predictions with those from simple LS/CMI categories highlights three main sources of misclassification that could be improved upon in future risk tool development, with some of these improvements being relatively straightforward to implement.

6.1 Items Are Not Equally Risk-Relevant

The first source of misclassification is the relative ability of items to differentiate between recidivists and non-recidivists. As shown in Figure 3, there is substantial variation in the importance of items, which led to misclassification for numerous offenders (see Figure 4). Typical cases of misclassification due to the differential risk relevance of items would be a) individuals having few important items and classified as low risk; or b) individuals with many less important items classified as high risk. In the first case, risk would be underestimated, and in the second one, it would be overestimated. Machine learning models, however, are able to mitigate these misclassifications by assigning greater weight to the few critical items that are strong predictors of recidivism, even when they appear in individuals with otherwise low overall risk scores. Conversely, it downplays less predictive items, ensuring that high-risk classifications are more accurately based on the most important factors rather than simply aggregating less relevant items.

The substantial variation in the risk relevance of LS/CMI items calls into question the weighting used in most criminological actuarial scales, wherein each item/risk factor is usually given a weight equal to one. If some items are more risk-relevant than others, they should potentially be given more weight. There are debates on the benefits of differentially weighting items, with some results indicating that complex combinations rarely outperform the simple summing of dichotomous items (Grann and Långström, 2007; Silver et al., 2000). However, differential weighting has its greatest impact when importance varies significantly across items (Ghiselli et al., 1981; Kline, 2016), which could be the case of the LS/CMI items in light of our results. In addition, recent studies of criminological risk scales based on the item-response theory (including some about the LS/CMI) found substantial variation in individual items' ability to discriminate between offender types. This also supports the need to weigh items accordingly (Brouillette-Alarie et al., 2023; Giguere and Lussier, 2016; Giguère et al., 2023). The three most predictive items according to our results (7, 7.1, and 8) are related to institutional misconducts and breach of probation or parole conditions. This is consistent with numerous studies in which, absent data on recidivism, use institutional misconducts as a proxy for potential recidivism. In more traditional predictive validity analyses based on a similar sample, items 7 and 8 of the LS/CMI were also found to be the most predictive (Giguère et al., 2023).

Recent methods like *FasterRisk* (Liu et al., 2022) offer a more sophisticated approach by using regularization to construct risk scores that optimize the weighting of variables based on their actual predictive value. Interestingly, when applying the developed algorithm on COMPAS, a risk assessment tool to predict future criminal misconduct (Northpointe, 2015), the authors find that by focusing on a subset of key variables (gender, age, and the number of prior offenses), they can achieve an AUC of approximately 0.73. This improvement is achieved by optimally weighting these variables and dividing the resulting scores into effective risk categories.

6.2 Risk-Relevant Information Collected by the LS/CMI is Not Taken Into Account in Risk Scores

Analyses related to the relative importance of items also reveal that among the 15 variables that most contributed to adequate classification, three were collected but not taken into account in LS/CMI total scores and risk categories. Namely, item 7.1 (number of institutional reports for problematic behavior), age, and item 1.2 (number of prior adult convictions) all enable improving the ability to discriminate recidivists from non-recidivists. Items 7.1 and 1.2 constitute continuous versions of items 7 and 1, which indicates that the loss of information resulting from their dichotomizing may reduce their predictive ability. This echoes the fact that conceptualizing risk factors as strictly present or absent, while practical, may limit the ability of assessors to adequately measure them and weight their relative importance in predicting recidivism.

6.3 Absence of Age Contributes to Misclassification

The other risk-relevant variable collected but not taken into account by LS/CMI risk scores is age, which not only has a relative importance, but also substantially contributes to misclassification, with the risk of older offenders being systematically overestimated. The absence of an age item in the LS/CMI is quite puzzling, as most criminological actuarial scales include an age item (e.g., VRAG-R, Static-99R/2002R). Moreover, the literature on the links between age and crime is quite exhaustive (Hirschi and Gottfredson, 1983). Over the past 10 years, actuarial scales, such as the Static-99 and the Static-2002, were revised in order to better take into account the effects of age. Whereas initial versions arbitrarily inflated the risk of young assessees, recent revisions conversely deflated the risk of those aged 35/40 or older (Helmus et al., 2012). Our results suggest that incorporating age into the LS/CMI would reduce the extent of misclassification of older individuals. Fréchette and Lussier (2022) similarly concluded that measurement errors in the LS/CMI were not random but rather directly linked to failure to account for the protective effect of ageing. Considering that other actuarial scales already account for age, it is advisable that the LS/CMI be amended accordingly.

A noteworthy "non-result" is that gender is not a major source of misclassification in the LS/CMI. There are ongoing debates in the risk assessment field concerning the appropriateness of current assessment tools for women. Because risk tools have mostly been devised for male detainees, they may not be well-suited for their particular needs (Belknap and Holsinger, 2006; Holtfreter and Morash, 2003). However, in terms of predictive validity, tools such as the LS/CMI have been found to perform just as well for men and women (Olver et al., 2014). Indeed, our results indicate that the LS/CMI is not inherently biased against women; gender is not found to exacerbate misclassification. This may be due to the fact that once all features (as well as their interactions) are accounted for, gender has a low predictive importance.[14]

6.4 ML Algorithms Enable Better Classification Than LS/CMI Total Scores and Risk Categories

Using ML algorithms to compute individual risk scores enables slight improvements in predictive validity, especially compared to the risk categories of the LS/CMI (Table 1).

[14] Note that the relative lack of gender bias applies to general purpose risk tools, not to those that focus explicitly on sexual offenses (Marshall et al., 2021).

Recent studies have compared traditional ways of computing risk scores (the "Burgess" method) to ML algorithms and found that the latter consistently outperformed the former (Duwe and Kim, 2016). In the study of Duwe and Kim (2016), improvements in predictive validity similar to those found in the current study (around +.03 to +.06) were enabled by using ML algorithms rather than using the Burgess method.

The question of whether these slight improvements in predictive validity warrant the loss in scoring transparency and the added implementation costs remains to be answered, but instruments such as the MnSTARR have already done so (Duwe, 2019), suggesting it is a realistic endeavor. Once the risk prediction formula is stabilized by the ML algorithm on a representative sample, it can easily be implemented in a scoring sheet or a web interface. Still, if one wishes to use categories, our algorithm in Section 5.5 shows that we may still do so while exploiting the predictive power of ML methods and avoid misclassification.

7 Limitations

The current study is not without limitations. First, the results reported here are limited to men and women incarcerated in Quebec's provincial prison system or serving a provincial community sentence in Quebec. Our sample is not necessarily representative of other populations. Specifically, it may not be generalizable to potentially higher-risk persons from Canadian federal prisons who have received a sentence of two years or more or to individuals from other provinces, countries, or ethnicities. Nonetheless, the methodology developed in the paper can easily be adapted to produce tailored categories for diverse populations.

Additionally, while this evaluation is conducted at the onset of incarceration and is not directly used by judges for pretrial release or in setting conditions of release, the LS/CMI scores may still influence treatment plans during incarceration. These plans could impact post-release supervision and, in turn, recidivism outcomes. It is important to recognize that the very assessment process can influence the outcomes it seeks to predict, as individuals with higher risk scores may receive more intensive interventions, potentially altering their likelihood of reoffending.

Another limitation of our approach is that the newly derived risk thresholds may be data-dependent. The improvements in classification accuracy we report may not extend to populations whose underlying distribution of risk factors differs from ours. If offender risk profiles vary across jurisdictions or change over time, static cutoffs derived from one dataset may lose relevance. However, the methodology we propose for defining risk categories is fully generalizable and can be applied to other populations. By re-estimating recidivism probabilities using specific populations, practitioners can optimize thresholds so as to better reflect their risk distribution.

Second, the follow-up period of the current study is limited to two years, which, while in line with field standards (Fazel and Wolf, 2015), may prove to be short and leave "little time" for at-risk individuals to reoffend.

Finally, the traditional practice of aggregating risk scores is likely to lead to misclassification in most assessment tools in addition to the LS/CMI. Yet, the extension of our results to other risk assessment tools remains to be validated.

8 Conclusion

Our study demonstrates the limitations of current actuarial tools used in predicting recidivism, which essentially rely on a summation of binary items. This approach may lead to significant misclassification because a given score may result from numerous combinations of individual items whose predictive power varies greatly. This highlights the need for a more nuanced method of devising risk categories. In particular, we find that older individuals are particularly susceptible to misclassification if age is not explicitly accounted for in the LS/CMI.

In order to minimize the extent of misclassification, we propose to use a random forest that uses all the variables included in the LS/CMI tool, thereby enhancing its predictive validity. We suggest a novel method for constructing risk categories that mitigates the misclassification biases. The proposed approach yields a higher predictive accuracy compared to those that rely on a simple aggregation of items. This not only ensures a more accurate representation of an offender's risk, but it also mitigates the social implications of erroneous classifications. Our recommendations likely apply to most assessment tools and should not be understood as specific to the LS/CMI, a risk tool that has solid conceptual foundations (Olver et al., 2014).

Our results suggest interesting avenues for future risk tool development:

1. Items should be weighted according to their ability to discriminate between recidivists and non-recidivists;

2. The arbitrary dichotomization of certain items unnecessarily removes relevant in-

formation and lowers the predictive validity of the score;

3. Tools should include the age of the offender to predict risk; and

4. ML algorithms could replace the traditional aggregation of items into actuarial scales, provided that the improvements in predictive validity warrants the implementation costs and loss in transparency.

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A Appendix-Additional Tables and Figures

Table A.1: Summary Statistics—LS/CMI Variables

Item Number	Risk Factor	Variable	Mean	Range	Included in Total Score	Type	Definition
1	Criminal History	CH_1	0.92	[0—1]	Yes	Binary	Any prior conviction
1.1	Criminal History	CH_1_1	0.86	[0—7]	No	Integer	Prior youth dispositions
1.2	Criminal History	CH_1_2	8.27	[0—21]	No	Integer	Adult convictions
2	Criminal History	CH_2	0.85	[0—1]	Yes	Binary	Two prior convictions
3	Criminal History	CH_3	0.78	[0—1]	Yes	Binary	Three prior convictions
4	Criminal History	CH_4	0.73	[0—1]	Yes	Binary	Three or more offenses for current sentence
5	Criminal History	CH_5	0.33	[0—1]	Yes	Binary	Arrested or charged before being 16 years old
6	Criminal History	CH_6	0.90	[0—1]	Yes	Binary	Ever incarcerated
7	Criminal History	CH_7	0.50	[0—1]	Yes	Binary	Institutional misconduct
7.1	Criminal History	CH_7_1	2.35	[0—10]	No	Integer	Number of behavior reports
8	Criminal History	CH_8	0.83	[0—1]	Yes	Binary	Breach of probation/parole conditions
9	Education/Employment	EE_9	0.66	[0—1]	Yes	Binary	Currently without a job
10	Education/Employment	EE_10	0.63	[0—1]	Yes	Binary	Frequently without a job
11	Education/Employment	EE_11	0.33	[0—1]	Yes	Binary	Never maintained a job for one year
12	Education/Employment	EE_12	0.41	[0—1]	Yes	Binary	Did not achieve grade 10
13	Education/Employment	EE_13	0.73	[0—1]	Yes	Binary	Did not achieve grade 12
14	Education/Employment	EE_14	0.54	[0—1]	Yes	Binary	Suspended/expelled from school
15	Education/Employment	EE_15	0.68	[0—3]	Yes	Binary	Not motivated/successful at school/work
16	Education/Employment	EE_16	0.73	[0—3]	Yes	Binary	Problematic relationships with peers at school/work
17	Education/Employment	EE_17	0.73	[0—3]	Yes	Binary	Problematic relationships with authority at school/work
18	Family/Marital	FM_18	1.56	[0—3]	Yes	Discrete	Not satisfied with intimate relationship or lack thereof
19	Family/Marital	FM_19	1.17	[0—3]	Yes	Discrete	Problematic relationship with parents
20	Family/Marital	FM_20	1.36	[0—3]	Yes	Discrete	Problematic relationship with other relatives
21	Family/Marital	FM_21	0.45	[0—1]	Yes	Discrete	A family member or spouse has a criminal record
22	Leisure/Recreation	LR_22	0.87	[0—1]	Yes	Binary	No prosocial activities
23	Leisure/Recreation	LR_23	1.01	[0—3]	Yes	Discrete	Has too much free time
24	Companions	CO_24	0.94	[0—1]	Yes	Binary	Links with criminalized individuals
25	Companions	CO_25	1.52	[0—3]	Yes	Discrete	Friends with criminalized individuals
26	Companions	CO_26	0.48	[0—1]	Yes	Binary	Few prosocial links
27	Companions	CO_27	1.01	[0—3]	Yes	Discrete	Few prosocial friends
28	Alcohol/Drug Problem	ADP_28	0.67	[0—1]	Yes	Binary	Ever had problems with alcohol consumption
29	Alcohol/Drug Problem	ADP_29	0.79	[0—1]	Yes	Binary	Ever had problems with drug consumption
30	Alcohol/Drug Problem	ADP_30	1.57	[0—3]	Yes	Discrete	Currently has problems with alcohol consumption
31	Alcohol/Drug Problem	ADP_31	1.41	[0—3]	Yes	Discrete	Currently has problems with drug consumption
32	Alcohol/Drug Problem	ADP_32	0.69	[0—1]	Yes	Binary	Criminal behavior related to alcohol/drug consumption
33	Alcohol/Drug Problem	ADP_33	0.48	[0—1]	Yes	Binary	Problem with spouse/family related to alcohol/drug consumption
34	Alcohol/Drug Problem	ADP_34	0.33	[0—1]	Yes	Binary	Problem at school/work related to alcohol/drug consumption
35	Alcohol/Drug Problem	ADP_35	0.21	[0—1]	Yes	Binary	Health problems related to alcohol/drug consumption
36	Procriminal Attitude/Orientation	PAO_36	1.35	[0—3]	Yes	Discrete	Favorable toward delinquency
37	Procriminal Attitude/Orientation	PAO_37	1.55	[0—3]	Yes	Discrete	Distrust toward society
38	Procriminal Attitude/Orientation	PAO_38	0.37	[0—1]	Yes	Binary	Resentful toward sentence/offense
39	Procriminal Attitude/Orientation	PAO_39	0.30	[0—1]	Yes	Binary	Uncooperative with supervision/treatment
40	Antisocial Pattern	AP_40	0.15	[0—1]	Yes	Binary	High-risk mental health problem
41	Antisocial Pattern	AP_41	0.45	[0—1]	Yes	Binary	Young and versatile delinquency
42	Antisocial Pattern	AP_42	0.71	[0—1]	Yes	Binary	Antisocial values
43	Antisocial Pattern	AP_43	0.63	[0—1]	Yes	Binary	Multiple problems

Notes: The table lists all the risk factors contained in the LS/CMI questionnaire (Andrews et al., 2000), including those not taken into account to output the total score. Binary items are yes/no questions that provide dichotomous responses. Item numbers that are integers each count for one point in the total score, with those initially rated on a scale of 4 binarized. Sub-items (i.e. item numbers 1.1, 1.2, 7.1) are ignored when computing the score.

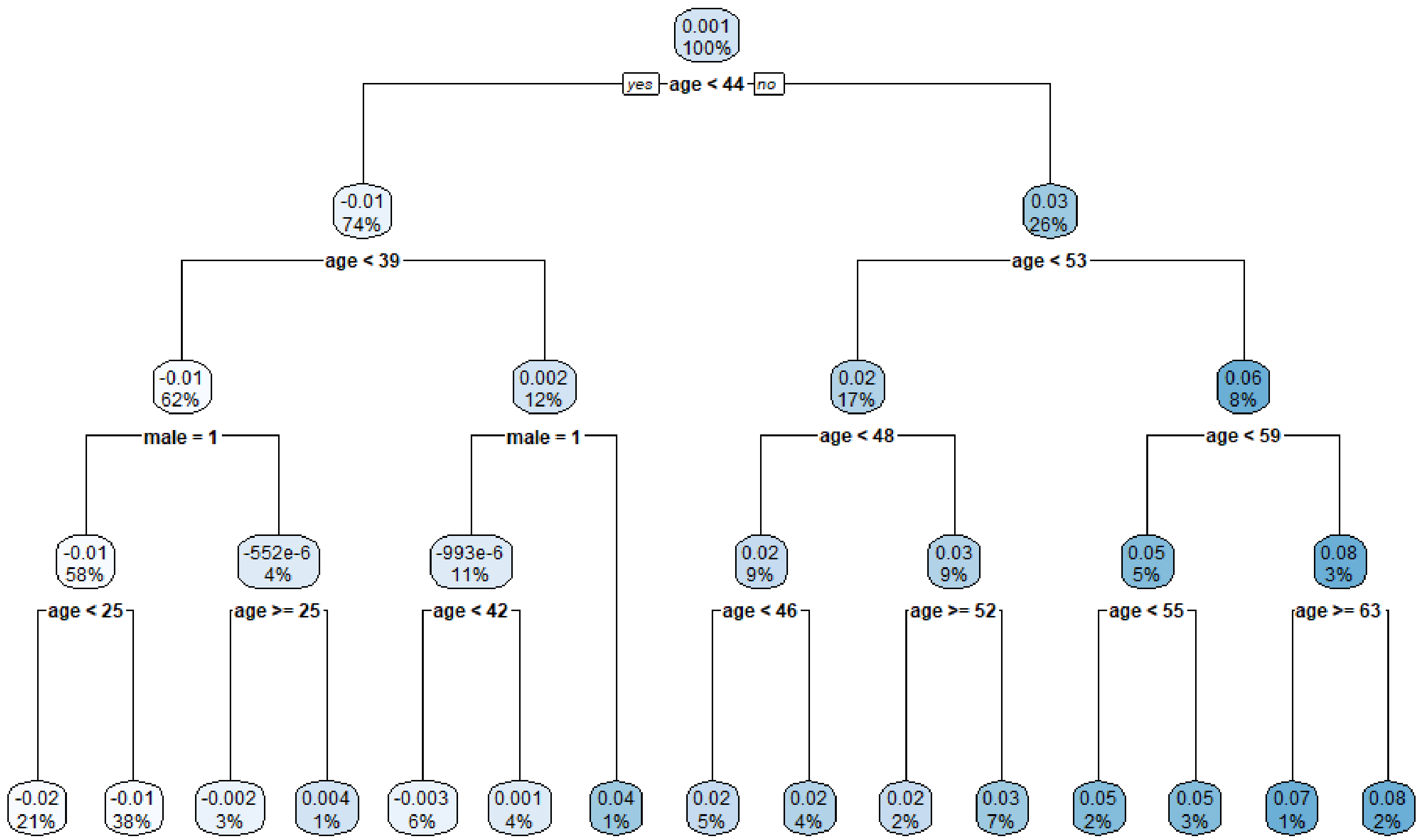

Figure A.1: Regression Tree: Which Characteristics Yield Errors in Predictions?

Notes: The error in prediction is defined as the difference between the probability given by the risk category and the probability obtained via ML. A positive error indicates that the risk category overestimates the risk of recidivism; in contrast, a negative error means an underestimation. The plot shows the regression tree obtained when predicting the error based on age and gender. Each node indicates the average error, as well as the proportion of observations contained within the node.

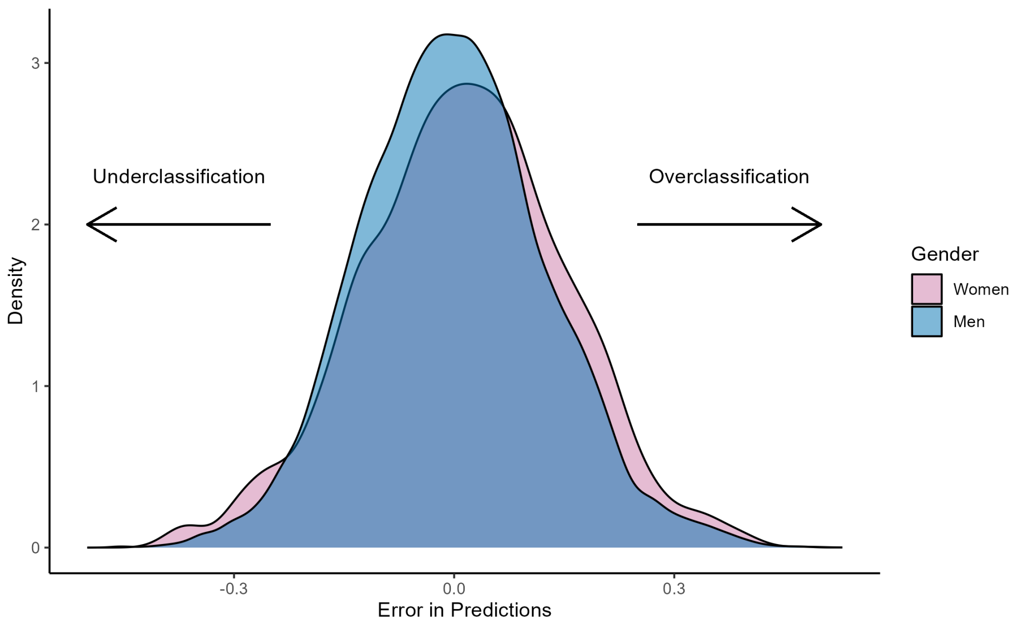

Figure A.2: Density of the Errors in Predictions by Gender

Notes: The error in prediction is defined as the difference between the probability given by the risk category and the probability obtained via ML. A positive error indicates that the risk category overestimates the risk of recidivism; in contrast, a negative error means an underestimation. The plot shows the distribution of the errors in the sample by gender.